Advanced International Macroeconomics B (EI136)

Class 2

Capital flows

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March 2, 2026

Outline of today class

- International capital flows.
  - Pattern and drivers (Milesi-Ferretti 2025, Agoraki et al 2024).
  - Drivers of volatility (Pagliari et al 2024).
  - Episodes of large flows (Forbes and Warnock 2021).
  - The "natural" level of capital flows (Burger et al. 2022).
- Digital assets and capital flows.
  - Bitcoin international flows (Cerutti, Chen and Hengge 2025).

CAPITAL FLOWS

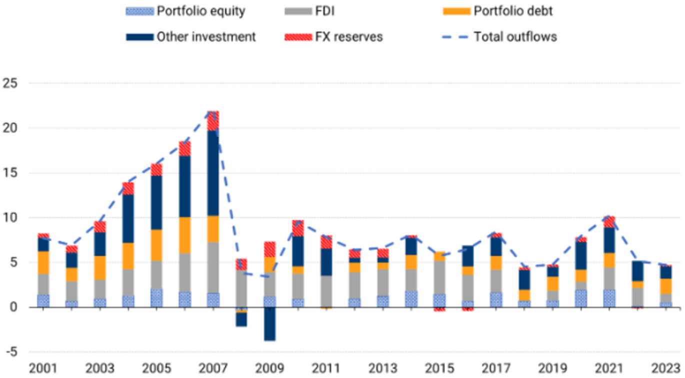
Capital flows: a changing pattern

- In 2008-2009 capital flows fell sharply, primarily for banking flows (Milessi-Ferretti 2025, BIS 2021).
  - Persistent retrenchment, with gross flows recovering only halfway to the pre-crisis peak.
  - Primarily in advanced economies. Flows have recovered in emerging markets (exception: emerging Europe).
- It is concentrated in "other" (banking) flows, especially in Advanced Europe (and emerging Europe).
  - Emerging markets' banks have increased their cross-border activity, at a regional level.

Flows by asset type

- Boom-bust cycle primarily in banking.

Figure 2. World Financial Outflows
Percent of world GDP



Milesi-Ferretti, Gian Maria (2025). "External Wealth of Nations complete update, 2023" Brookings

Drivers of flows to emerging economies

- Survey of the literature (Koepke 2019), with focus on gross inflows.
- Heterogeneous situation depending on the type of flows.
  - Global (push) factors matter more for portfolio equity and portfolio debt.
  - Push factors matter less for "other" inflows (mostly bank loans). This can be sensitive to the specific push factors.
  - Local (pull) factors matter more for bank inflows.

Summary of drivers

- Global risk and interest rates matter, as does local risk.

Type	Driver	Portfolio Equity	Portfolio Debt	Banking Flows
Push	Global risk aversion	−	−	−
	Mature economy interest rates	−	−	-
	Mature economy output growth	+	+	?
Pull	Domestic output growth	+	+	+
	Asset return indicators	+	+	+
	Country risk indicators	-	-	−

+	Strong evidence for positive relationship
+	Some evidence for positive relationship
?	Mixed evidence, no clear relationship
-	Some evidence for negative relationship
−	Strong evidence for negative relationship

Figure 1. Drivers of EM Capital Flows by Major Component. [Colour figure can be viewed at wileyonlinelibrary.com]

Notes: The matrix summarizes the available evidence on the role of push and pull factors for the major capital flows components. For example, the red cell in the top left corner of the matrix indicates that there is strong evidence that an increase in global risk aversion leads to a reduction in portfolio equity flows to emerging markets.

Koepke, Robin (2019). "What drives Capital Flows to Emerging Markets: A Survey of the Empirical Literature", *Journal of Economic Surveys* 33(2), pp. 516-540.

Risk as a driver of flows

- How sensitive are capital flows to various types of global risk (Agoraki et al 2024)?
  - Geopolitical risk (GPR).
  - Economic policy uncertainty (EPU).
  - World uncertainty index (WUI).
- Capital flows of investment funds (EPFR), distinguishing between various types of funds.
- Different risk indices have different impacts.
- No impact when considering overall capital flows, including FDI and banking flows.

Contrasted effects

- Higher GPR reduces all capital flows, except more passive ETF (retrenchment to domestic market).
- High EPU reduces equity flows (especially if high GPR), with no effect on bonds flows.
- High WUI reduces equity flows, but raises bonds flows (flight to safety).

Panel A: Geopolitical Risk Index						
	(1) MF_equity	(2) MF_institutional	(3) MF_retail	(4) MF_ETF	(5) MF_bond	(6) MF_bond_ETF
GPR	−0.054*	−0.196***	0.010	0.153***	−0.255***	−0.122***
	[-2.08]	[-6.18]	[0.40]	[5.44]	[-8.13]	[-4.18]
Panel B: Economic Policy Uncertainty Index						
	(1) MF_equity	(2) MF_institutional	(3) MF_retail	(4) MF_ETF	(5) MF_bond	(6) MF_bond_ETF
EPU	−0.196***	−0.229***	−0.245***	−0.067**	−0.032	−0.465***
	[-6.90]	[-7.46]	[-7.68]	[-2.63]	[-0.90]	[-4.20]
Panel C: World Uncertainty Index						
	(1) MF_equity	(2) MF_institutional	(3) MF_retail	(4) MF_ETF	(5) MF_bond	(6) MF_bond_ETF
WUI	−0.095***	−0.154***	−0.112***	0.001	0.044**	−0.157***
	[-4.88]	[-6.85]	[-6.73]	[0.08]	[2.68]	[-4.85]

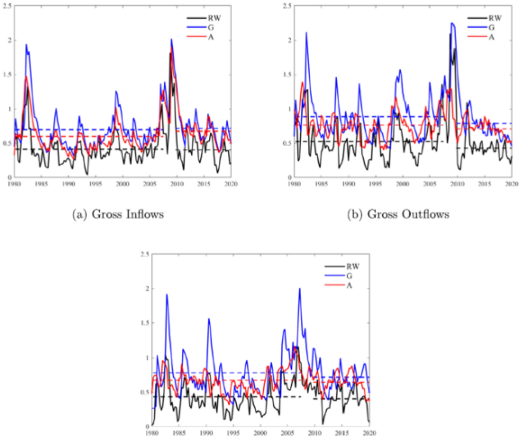
Agoraki, Maria-Eleni K., Haoran Wu, Tongbin Xu, and Min Yang (2024). "Money never sleeps: Capital flows under global risk and uncertainty, *Journal of International Money and Finance* 141.

Drivers of volatility

- Instead of using the level of flows as the dependent variables, use a measure of their volatility (Pagliari et al 2024).
- Consider various measures, focusing on emerging and developing economies (33).
    - Volatility of the residuals from a ARIMA univariate regression of the quarter-to-quarter change in flows (using GARCH if needed).
    - Rolling window standard deviation of the deviation of flows from average (RW).
    - Standard deviation from a GARCH(1,1) regression of the quarter-to-quarter change in flows.
- RW estimates tend to be the lowest, ARIMA estimates tend to be in between the other two.
- Reduction in volatility of outflows since 2008, but not of inflows.
    - Small increase in FDI volatility, decrease in portfolio flows volatility, while banking flows volatility remained steady.

Pattern of volatility

- Decrease in outflows volatility.



(a) Gross Inflows

(b) Gross Outflows

(c) Net Inflows

Fig. 1. Volatility Estimates for Aggregate Flows in EMDEs. *Notes:* Dotted lines refer to pre- and post-GFC averages. RW: rolling-window standard deviation; G: GARCH(1,1); A: ARIMA(p,1,q).

Pagliari, Maria Sole, and Swarnali Ahmed Hannan (2024). "The volatility of capital flows in emerging markets: Measures and determinants" *Journal of International Money and finance* 145

Panel analysis

- Regress volatility on push and pull factors.
  - Push: US interest rate (shadow), US growth, stick market volatility.
  - Pull: domestic interest rate, inflation, growth, exchange rate.
- Heterogeneous effects across drivers and categories.
- Inflow volatility (ARIMA results).
  - Higher US interest rate raises it for FDI.
  - Higher US stock market volatility has a positive impact overall and for FDI, but negative for portfolio.
  - Higher domestic inflation raises volatility for FDI and banks.
  - Effects somewhat larger after 2007.
- Outflow volatility (ARIMA results).
  - Higher US interest rate raises it for FDI.
  - Higher US stock market volatility has a positive impact overall and for banks.

Estimates for inflows volatility

- Heterogeneous, role of global variables.

Table 3a
Sign effects of push and pull factors - volatility of gross inflows.

	Total	FDI	Portfolio	Port. Debt	Port. Equity	Other	Banks	Official	Non-banks	Total private
ARIMA										
US shadow rate		+								
Realised returns volatility	+	+	-	-				-		
US Real GDP										
US CPI							+		+	
US M2										
US Current Account			-	-				+		
Real GDP					-					
CPI		+			-	+				+
Short-term int. rate				-	+				+	
Exchange rate vs USD							+			
RW										
US shadow rate		+							+	
Realised returns volatility	+						+			
US Real GDP			-			-	-	-		
US CPI										
US M2		+	-					-		
US Current Account								+		
Real GDP										
CPI										
Short-term int. rate									+	
Exchange rate vs USD							+	+		

Notes: Results based on the estimation of Equation (7), where ARIMA measures of volatility for different functional categories are the dependent variables, while the set of regressors is reported in the first column + positive and significant (95% in red; 90% in blue) impact; - negative and significant (95% in red; 90% in blue) impact.

Pagliari, Maria Sole, and Swarnali Ahmed Hannan (2024). "The volatility of capital flows in emerging markets: Measures and determinants" *Journal of International Money and finance* 145

Estimates for outflows volatility

Table 3b

Sign effects of push and pull factors - volatility of gross outflows.

	Total	FDI	Portfolio	Port. Debt	Port. Equity	Other	Banks	Official	Non-banks	Total private
ARIMA										
US shadow rate										
Realised returns volatility										
US Real GDP		-								
US CPI							+	+		
US M2									+	
US Current Account										
Real GDP		-		-						
CPI			-	-					-	
Short-term int. rate										
Exchange rate vs USD										
RW										
US shadow rate		+								
Realised returns volatility										
US Real GDP										
US CPI							+	+		
US M2										
US Current Account			+	+	+					
Real GDP	-	-								
CPI						-			-	-
Short-term int. rate										
Exchange rate vs USD										

Notes: Results based on the estimation of Equation (7), where ARIMA measures of volatility for different functional categories are the dependent variables, while the set of regressors is reported in the first column. + positive and significant (95% in red; 90% in blue) impact; - negative and significant (95% in red; 90% in blue) impact.

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EPISODES OF LARGE CAPITAL FLOWS

Large movements in flows

- Episodes of large movements in capital flows are more relevant for policy than times of usual flows (large movements are harder to absorb).
- Forbes and Warnock (2021) focus on gross flows. Four types of episodes:
  - *Surge*: large positive gross inflows.
  - *Stop*: large negative gross inflows.
  - *Flight*: large positive gross outflows.
  - *Retrenchment*: large negative gross outflows.
- What is large: unusual in recent historical perspective.
  - $C_t = Flows_t + ... + Flows_{t-3}$: sum of flows over 4 quarter (smooth things out) of private flows.
  - $\Delta C_t = C_t - C_{t-4}$: change from one year to the next (non-overlapping).
  - Compute rolling mean and standard deviation over previous 5 years. Large if ΔC_t is one stdev above the rolling mean.
- Episodes are frequent and come in waves.

Incidence of episodes

- All episodes come in waves.

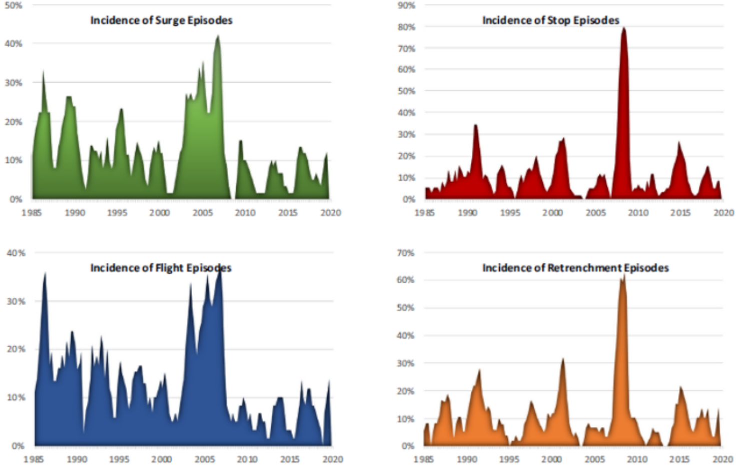

Fig. 1a. Incidence of Surges, Stops, Flight and Retrenchment: Full Sample. **Notes:** Share of all countries in sample experiencing each type of capital flow episode each quarter from 1985q1-2020Q2. Sample includes at most 58 countries. See Appendix Table A1 for details on dates for capital flow data by country.

Forbes, Kristin, and Francis Warnock (2021). "Capital Flow Waves—or Ripples? Extreme Capital Flow Movements Since the Crisis" *Journal of International Money and Finance* 116.

What drives episodes?

- Regress a dummy variable on global, regional, and local factors (complementary log for extreme events, cloglog).
- Global factors.
  - VXO stock market volatility, broad money aggregate of advanced countries, interest rate in advanced countries, world growth.
- Regional factors (contagion).
  - Episodes in other countries in the region, episodes in neighboring countries weighted by trade, or by banking, linkages.
- Local factors.
  - Financial sector depth, capital controls, debt to GDP, growth.
- Global risk, interest rates and growth matters for all episodes. Regional dimension is important.
- Effect of global factors observed pre-2007, no significant effect since. .

Drivers of episodes: entire sample

- Global and regional factors matter.

Table 4
Regression Results with Global, Contagion and Local Variables, new Baseline.

	Full Sample (1986–2020)				Pre-Crisis (1986–2007)				Post-Crisis (2010–2020)			
Global Vars	Surge (1)	Stop (2)	Flight (3)	Retrench (4)	Surge (5)	Stop (6)	Flight (7)	Retrench (8)	Surge (9)	Stop (10)	Flight (11)	Retrench (12)
Risk	-0.039^{**}	0.033^{**}	-0.030^{**}	0.029^{**}	-0.042^{**}	0.021	-0.042^{**}	0.033^{**}	−0.028	0.006	−0.036	0.000
	(0.008)	(0.004)	(0.009)	(0.005)	(0.010)	(0.013)	(0.008)	(0.013)	(0.024)	(0.028)	(0.029)	(0.024)
Liquidity	0.019^{**}	-0.020^{**}	0.017^{**}	−0.003	0.023^{**}	-0.040^{**}	0.017^{**}	-0.021^{**}	−0.006	−0.028	0.012	0.023
	(0.008)	(0.006)	(0.007)	(0.007)	(0.009)	(0.009)	(0.007)	(0.010)	(0.027)	(0.026)	(0.029)	(0.032)
Monetary Policy	0.114^{**}	0.129^{**}	0.123^{**}	0.100^{**}	−0.027	0.178^{**}	−0.059	0.121^{**}	0.050	0.380^{**}	−0.015	0.512^{**}
	(0.020)	(0.019)	(0.020)	(0.020)	(0.047)	(0.041)	(0.043)	(0.043)	(0.149)	(0.185)	(0.161)	(0.201)
Growth	0.134^{**}	-0.187^{**}	0.112^{**}	-0.203^{**}	0.160^{**}	-0.209^{*}	0.206^{**}	-0.272^{**}	-0.306^{**}	0.048	−0.220	0.025
	(0.050)	(0.044)	(0.055)	(0.049)	(0.078)	(0.108)	(0.069)	(0.106)	(0.129)	(0.129)	(0.137)	(0.134)
Oil Prices	0.001	0.002	−0.002	0.003^{*}	−0.001	−0.003	-0.006^{**}	−0.000	0.011^{**}	-0.016^{**}	0.007^{*}	-0.019^{**}
	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.002)	(0.003)	(0.004)	(0.005)	(0.004)	(0.006)
Regional Contagion	0.422^{**}	0.610^{**}	0.229^{*}	0.530^{**}	0.500^{**}	0.413^{**}	0.055	0.421^{**}	0.065	0.322	−0.014	0.294
	(0.146)	(0.145)	(0.130)	(0.138)	(0.214)	(0.200)	(0.146)	(0.183)	(0.256)	(0.299)	(0.280)	(0.273)
Domestic Var												
GDP Growth	0.080^{**}	-0.062^{**}	0.038^{**}	-0.041^{**}	0.055^{**}	-0.102^{**}	0.019	-0.044^{**}	0.123^{**}	−0.001	0.024	−0.001
	(0.013)	(0.017)	(0.012)	(0.015)	(0.015)	(0.020)	(0.015)	(0.019)	(0.034)	(0.040)	(0.036)	(0.033)
Obs.	**5787**	**5787**	**5787**	**5787**	**2956**	**2956**	**2956**	**2956**	**2384**	**2384**	**2384**	**2384**

Forbes, Kristin, and Francis Warnock (2021). "Capital Flow Waves—or Ripples? Extreme Capital Flow Movements Since the Crisis" *Journal of International Money and Finance* 116.

Splitting the sample

- Before the 2008 crisis, some global factors matter, as regional ones.
- No effect of global factors after that.

Table 4
Regression Results with Global, Contagion and Local Variables, new Baseline.

	Full Sample (1986–2020)				Pre-Crisis (1986–2007)				Post-Crisis (2010–2020)			
Global Vars	Surge (1)	Stop (2)	Flight (3)	Retrench (4)	Surge (5)	Stop (6)	Flight (7)	Retrench (8)	Surge (9)	Stop (10)	Flight (11)	Retrench (12)
Risk	−0.039**	0.033**	−0.030**	0.029**	−0.042**	0.021	−0.042**	0.033**	−0.028	0.006	−0.036	0.000
	(0.008)	(0.004)	(0.009)	(0.005)	(0.010)	(0.013)	(0.008)	(0.013)	(0.024)	(0.028)	(0.029)	(0.024)
Liquidity	0.019**	−0.020**	0.017**	−0.003	0.023**	−0.040**	0.017**	−0.021**	−0.006	−0.028	0.012	0.023
	(0.008)	(0.006)	(0.007)	(0.007)	(0.009)	(0.009)	(0.007)	(0.010)	(0.027)	(0.026)	(0.029)	(0.032)
Monetary	0.114**	0.129**	0.123**	0.100**	−0.027	0.178**	−0.059	0.121**	0.050	0.380**	−0.015	0.512**
Policy	(0.020)	(0.019)	(0.020)	(0.020)	(0.047)	(0.041)	(0.043)	(0.043)	(0.149)	(0.185)	(0.161)	(0.201)
Growth	0.134**	−0.187**	0.112**	−0.203**	0.160**	−0.209*	0.206**	−0.272**	−0.306**	0.048	−0.220	0.025
	(0.050)	(0.044)	(0.055)	(0.049)	(0.078)	(0.108)	(0.069)	(0.106)	(0.129)	(0.129)	(0.137)	(0.134)
Oil Prices	0.001	0.002	−0.002	0.003*	−0.001	−0.003	−0.006**	−0.000	0.011**	−0.016**	0.007*	−0.019**
	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.003)	(0.002)	(0.003)	(0.004)	(0.005)	(0.004)	(0.006)
Regional	0.422**	0.610**	0.229*	0.530**	0.500**	0.413**	0.055	0.421**	0.065	0.322	−0.014	0.294
Contagion	(0.146)	(0.145)	(0.130)	(0.138)	(0.214)	(0.200)	(0.146)	(0.183)	(0.256)	(0.299)	(0.280)	(0.273)
Domestic Var												
GDP	0.080**	−0.062**	0.038**	−0.041**	0.055**	−0.102**	0.019	−0.044**	0.123**	−0.001	0.024	−0.001
Growth	(0.013)	(0.017)	(0.012)	(0.015)	(0.015)	(0.020)	(0.015)	(0.019)	(0.034)	(0.040)	(0.036)	(0.033)
Obs.	**5787**	**5787**	**5787**	**5787**	**2956**	**2956**	**2956**	**2956**	**2384**	**2384**	**2384**	**2384**

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BENCHMARKING CAPITAL FLOWS

"Natural" capital flows

- Natural level of variables is common in macroeconomics: natural real interest rate r_t^*, or output level to compute output gap, $y_t - y_t^*$.
- What is the equivalent for capital flows, KF_t^* (Burger, Warnock and Warnock 2022).
- Compute the capital inflows in a destination country, $KF_{d,t}^*$, if the actual savings of the rest of the world, $S_{ROW,t}$, are allocated using the previous share of country d in the ROW portfolio of bonds and equity, $\omega_{ROW,d,t-i}$ (average over the last 5 years, constant shares distort the results).

$$KF_{d,t}^* = \left(\frac{1}{5} \sum_{i=1}^{5} \omega_{ROW,d,t-i} \right) S_{ROW,t}$$

- Theoretical underpinning: along a steady growth path, the portfolio allocation is steady.
- Important to allow for the portfolio shares to evolve through time.

- Actual capital flows and $KF^*_{d,t}$.

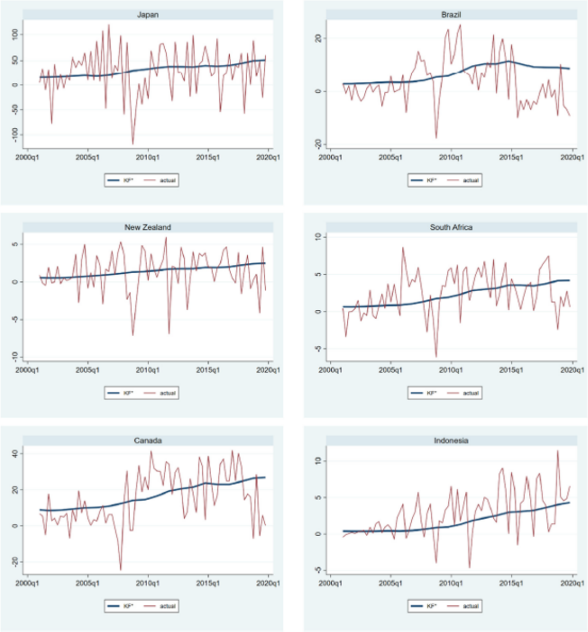


Fig. 2. KF* and Gross Portfolio Inflows (2000q4–2019q4, billions of USD).
The graphs show quarterly portfolio inflows (the thin volatile line) and KF* (the slow-moving thick blue line).

Burger, John, Francis Warnock, and Veronica Cacdac Warnock (2022). "A Natural Level of Capital Flows", *Journal of Monetary Economics* 130, pp. 1-16.

Use of benchmark capital flows

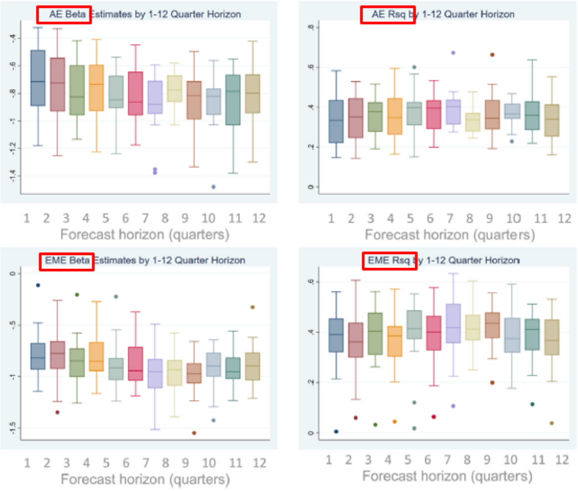
- Do actual capital flows revert to the natural level? Regress future change of flows (h periods ahead) on gap:

$$flows_{i,t+h} - flows_{i,t} = \alpha_{i,h} + \beta_{i,h} \left(flows_{i,t} - KF^{*}_{i,t} \right) + \varepsilon_{i,t}$$

- Reversion occurs if $\beta_{i,h} < 0$, with full at $\beta_{i,h} = -1$.
- Evidence of significant reversion, with large R2.
- Gap from KF^{*} helps forecast flows.
  - Positive gap raises the likelihood of a sudden stop.
  - Positive gap associated with larger contraction during the global financial crisis and Covid.

Reversion of capital flows

- Positive gap followed by reduction in inflows.



Burger, John, Francis Warnock, and Veronica Cacdac Warnock (2022). "A Natural Level of Capital Flows", *Journal of Monetary Economics* 130, pp. 1-16.

- Positive inflow gap raises the likelihood of sudden stop.

Table 1
*KF** and extreme capital flow episodes.

Panel A	Sudden Stops	Surges
*KF*gap*	16.109***	−0.532
	(4.984)	(3.064)
Global Variables		
Global GDP Growth	0.541***	−0.178*
	(0.208)	(0.103)
Risk	0.067***	−0.003
	(0.021)	(0.076)
Liquidity	0.115	−0.002
	(0.070)	(0.041)
Oil Prices	−0.004	0.003
	(0.003)	(0.004)
Monetary Policy	0.286	0.213
	(0.187)	(0.141)
Local and Contagion Variables		
Local GDP Growth	0.022	0.065***
	(0.028)	(0.024)
Regional Contagion	−0.129	0.219
	(0.167)	(0.167)
Observations	1783	1783
Countries	26	26

Panel B	**Prob (Stop) $t+6$ quarters**
$KF^*gap = 0\%$	8.4%
$KF^*gap = 3.4\%$	14.1%
$KF^*gap = 6.8\%$	23.1%
$KF^*gap = 3.4\%$ & Global growth = 4.2%	30.7%

Burger, John, Francis Warnock, and Veronica Cacdac Warnock (2022). "A Natural Level of Capital Flows", *Journal of Monetary Economics* 130, pp. 1-16.

Demographics

- World population is aging, even in emerging countries.
    - Two dimensions: lower fertility rates, and rising life expectancy.
- Impact on capital flows using a OLG model with young - middle aged - old (Barany, Coeurdacier, Guibaud, 2023).
- Countries have heterogeneous financial and policy frameworks:
    - Borrowing constraint for young agents, affects borrowing demand (tighter in emerging economies).
    - "Pay as you go" retirement system, where middle aged agents pay for current old one, affects savings (more generous in advanced economies).

Offsetting patterns

- Global aging leads to capital flows from emerging to advanced countries.
  - Higher savings. Limited ability to invest domestically in emerging economies (constraint). So save in advanced markets.
- Differentiated aging leads capital to flows to younger countries (emerging ones).
- Falling fertility is the component of aging that has the most effect, and amplifies the impact of life expectancy.
- Good fit of capital flows from a rich model and the data.

DIGITAL CAPITAL FLOWS

Flows in Bitcoin

- Digital currencies are gaining prominence. What is their presence in capital flows, and what drives them (Cerutti, Chen and Hengge 2025)?
- Construction of data is already a challenge.
  - No clear residency concept, unlike the balance of payments.
- Construct estimates using three approaches.
  - On chain transaction from cross-exchange flows.
  - On chain transaction using wallet information and allocation of exchange flows based on wallet composition (Chainanalysis).
  - Off chain (via a third party) based on transaction from bitcoin to fiat currency, matching close currency-bitcoin-currency transactions.
- Standard capital flows estimates: EPFR (mutual funds) and Institute of International Finance (portfolio flows).

- Three approaches, with pros and cons.

Table 5
Overview of bitcoin cross-border flows.

	Blockchain	Chainalysis	LocalBitcoins
Type	On chain	On chain	Off chain
Key assumptions		Web traffic = residency	Fiat currency = residency
Pros	Transaction-level data	Many exchanges	Transaction-level data
		Representative of on-chain volume	Relatively long sample
Cons	No country mapping	Imprecise for VPN use	Not representative of off-chain volume
		Web traffic assumption simplified	Imprecise for dominant currencies

This table provides an overview of the Bitcoin cross-border flows datasets, including the key assumptions to identify the residency of market participants and their pros and cons.

Cerutti, Eugenio, Jiagian Chen, and Martina Hengge (2025). "A primer on bitcoin cross-border flows: Measurement and drivers", *Journal of International Money and Finance* 159.

Pattern of flows

- Broad geographical use of Bitcoing flows. They are inversely related to usual flows. Suggests a role to get around capital controls.

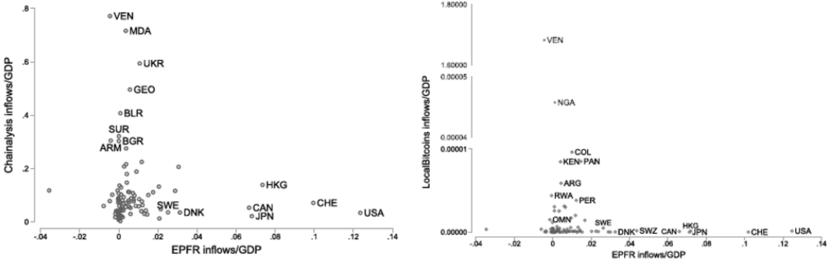


Fig. 9. Bitcoin and EPFR inflows. The LHS figure shows the average of 2019–2022 monthly cross-border Chainalysis inflows as percent of (2017–2022) average annual GDP and the average of 2017–2022 monthly EPFR inflows as percent of average annual GDP. The RHS figure shows the average of 2017–2022 monthly cross-border LocalBitcoins inflows as percent of average annual GDP and the average of 2017–2022 monthly EPFR inflows as percent of average annual GDP.

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Panel analysis

- Regress flows on push and pull factors.
  - Distinguish on chain (Chainanalysis) and off chain flows.
  - Push: standard macro (Vix, dollar broad index) and crypto factor (fear & greed sentiment, with higher values more positive towards crypto).
  - Pull: standard macro (inflation, interest differential) and crypto factor (gap between bitcoin-based exchange rate agains $ and standard exchange rate).
- Panel with only macro factors.
  - Push matter for standard flows, but only dollar broad dollar index matters for on chain crypto.
- Include crypto factors.
  - On chain crypto flows sensitive to fear & greed, inversely related to Vix (move to non-standard risky assets).
  - Off chain crypto flows sensitive to pull conditions.
- Results driven by emerging economies.

Panel with standard factors

- Push matter much less for crypto flows.

Table 8
Traditional global and domestic drivers of Bitcoin cross-border flows and capital flows.

	Bitcoin flows				Capital flows			
	(1) CA in	(2) CA out	(3) LB in	(4) LB out	(5) EPFR bond in	(6) EPFR equity in	(7) IIF debt in	(8) IIF equity in
Global drivers								
VIX	0.47	0.48	0.02	0.03	−8.52***	−2.09***	−1.99***	−3.05***
	[0.35]	[0.36]	[0.10]	[0.23]	[2.41]	[0.66]	[0.32]	[1.00]
Broad dollar	−2.20**	−2.15**	−0.01	−0.25	−1.72	−5.25***	−2.24*	−0.68
	[0.86]	[0.88]	[0.23]	[0.64]	[1.46]	[1.64]	[1.14]	[2.87]
Domestic drivers								
Inflation	−0.02	−0.03	−0.29	−1.74	−0.41	−0.32*	0.04	−0.67***
	[0.02]	[0.03]	[0.48]	[1.07]	[0.38]	[0.17]	[0.21]	[0.16]
Interest differential	0.07	0.04	−1.74***	−0.33	0.96	0.76	−2.18	3.75
	[0.36]	[0.40]	[0.55]	[0.44]	[0.87]	[0.74]	[2.96]	[3.07]
Observations	1741	1741	2098	2098	2630	2630	917	1159
R2	0.86	0.85	0.93	0.70	0.37	0.47	0.25	0.20

Robust standard errors in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Notes: We scale monthly flows by country-level average annual GDP. Next, we divide flows-to-GDP by their standard deviation. The dependent variable is the resulting ratio scaled by 100. CA in columns 1–2 refers to Chainalysis while LB in columns 3–4 refers to LocalBitcoins. All columns control for the lagged dependent variable and country-year fixed effects. All regressions are estimated with double clustered standard errors at the country and month level.

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Panel with standard and crypto factors

- Role of global sentiment breaks impact of broad dollar for crypto.

Table 9
Traditional and bitcoin-specific global and domestic drivers of Bitcoin cross-border flows and capital flows.

	Bitcoin flows				Capital flows			
	(1) CA in	(2) CA out	(3) LB in	(4) LB out	(5) EPFR bond in	(6) EPFR equity in	(7) IIF debt in	(8) IIF equity in
Global drivers								
VIX	0.66***	0.67***	0.02	0.09	−8.98***	−2.52***	−2.26***	−2.30*
	[0.22]	[0.23]	[0.07]	[0.23]	[2.55]	[0.81]	[0.72]	[1.10]
Broad dollar	−0.55	−0.54	0.41	0.48	−0.52	−5.67***	−2.55*	−1.45
	[1.05]	[1.09]	[0.32]	[0.77]	[1.92]	[1.69]	[1.25]	[4.30]
Crypto fear&greed	0.54***	0.52**	0.04	0.14	0.28	0.27	0.04	0.77**
	[0.19]	[0.20]	[0.03]	[0.15]	[0.30]	[0.22]	[0.32]	[0.32]
Domestic drivers								
Inflation	−0.61	−0.58	−2.22	−3.62*	−9.98*	−5.42*	−6.11	2.17
	[0.86]	[0.84]	[1.57]	[1.88]	[5.38]	[2.72]	[4.37]	[5.14]
Interest differential	0.91	0.90	2.53**	0.65	3.32	2.37	3.05	1.67
	[0.62]	[0.63]	[1.23]	[0.85]	[2.34]	[1.70]	[4.25]	[5.94]
BTC parallel premium	−0.04	−0.04	−0.01	0.06**	−0.03	0.02	−0.48	−0.05**
	[0.11]	[0.12]	[0.01]	[0.03]	[0.02]	[0.02]	[0.38]	[0.02]
Observations	929	929	1253	1253	1221	1221	531	629
R2	0.88	0.87	0.93	0.80	0.33	0.40	0.24	0.21

Robust standard errors in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Notes: We scale monthly flows by country-level average annual GDP. Next, we divide flows-to-GDP by their standard deviation. The dependent variable is the resulting ratio scaled by 100. CA in columns 1–2 refers to Chainalysis while LB in columns 3–4 refers to LocalBitcoins. All columns control for the lagged dependent variable and country-year fixed effects. All regressions are estimated with double clustered standard errors at the country and month level.

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